ENGINEERING SCIENCE

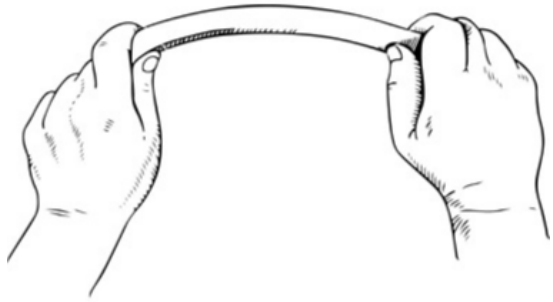
Higher

Revised May 2020

Mechanisms and Structures

Unit 3.4 Materials

1. Properties of materials
2. Testing
3. Load v Extension graphs, Strain and Stress, stress v strain graphs
4. Youngs modulus
5. Factors of Safety
6. Past paper questions



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Class: _____

Teacher: _____

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Outcomes and Assessment Standards

This unit will build on skills learned in National 5 Engineering Science in order to further deepen your understanding of Mechanisms and Structures to enable you to design your own working systems.

To make sure you have all the understanding needed to succeed at Higher Engineering Science, record your progress using the following colour codes:

Red: I need to work on this topic further to understand it.

Amber: I need a bit of help to fully understand this topic.

Green: I am confident in my understanding of this topic.

I can research and investigate a range of complex mechanisms and structures by:

- Using the systems approach to analyse mechanisms and structures.
- Describing or producing diagrams of a range of complex structures.
- Describing or producing diagrams of a range of complex mechanisms.
- Investigating the properties of a range of materials used in mechanisms.

I can develop mechanical or structural solutions to solve complex problems by:

- ☐ Identifying key aspects of the problem.
- ☐ Applying knowledge and understanding of structures, materials and/ or mechanisms.
- ☐ Carrying out calculations to assist the selection of materials or component sizes.
- ☐ Describe the properties of a material from test data.
- ☐ Identify important points on the graph.
- ☐ Describe the effect of increased loading on a test piece.
- ☐ Calculate Young's Modulus, stress and strain.

Properties of materials

It is not only the shape of a structure that will influence its overall performance but also the material that each member in the structure is made from. If any one member was to fail within the structure itself, it would create a domino effect on the other members and ultimately the structure would collapse.

In order to select a material for a particular purpose the structural engineer must consider a range of materials, all with different properties. The structural engineer must research and consider a range of different types of materials, or modify materials in specific ways to improve the performance, durability and cost effectiveness of the structure to ensure the choice of material that is selected is best suited to the job in hand. It is important for you to explain material properties using key terminology.

The most common properties to be considered include:

1. **STRENGTH** - the ability of a material to resist force. All materials have some degree of strength - the greater the force the material can resist, the stronger the material. Some materials can be strong in tension but weak in compression, for example mild steel. The converse can also be true, as is the case with concrete, which is strong in compression but weak in tension. Hence, the reason that concrete is often reinforced with mild steel.

2. **ELASTICITY** - the ability of a material to return to its original shape or length once an applied load or force has been removed. A material such as rubber is described as elastic because it can be stretched but when it is released it will return to its original condition.

3. **PLASTICITY** - the ability of a material to change its shape or length under a load and stay deformed even when the load is removed. Think of pressing your thumb into clay as it has a high plasticity.

4. **DUCTILITY** - the ability of a material to be bent and stretched without fracturing and be formed into shapes such as very thin sheets or very thin wire. Copper, for example, is very ductile and behaves in a plastic manner when stretched.

5. **BRITTLENESS** - the property of being easily cracked, snapped or broken. It is the opposite of ductility and therefore the material has little plasticity and will fail under loading without stretching or changing shape. Cast iron and glass are obvious examples of materials that are brittle.

6. **MALLEABILITY** - the ability of a material to be shaped, worked or formed without fracturing. It is closely related to the property of plasticity. An example of a malleable material is gold, where it can be shaped into multiple objects. It is very likely that materials will have multiple properties. Many materials will be mixed or coated with another material or to ensure that a particular material requirement is achieved. For example, steel is strong but it corrodes easily due to its iron content which easily rusts when exposed to moisture. To stop this corrosion occurring by covering it in particular paints.

7. **TOUGHNESS** - the ability to absorb a sudden sharp load without causing permanent deformation or failure. Tough materials require high elasticity.

8. **HARDNESS** - the ability to resist erosion or surface wear. Hard materials are used in situations where two surfaces are moving across or over each other.

Material testing

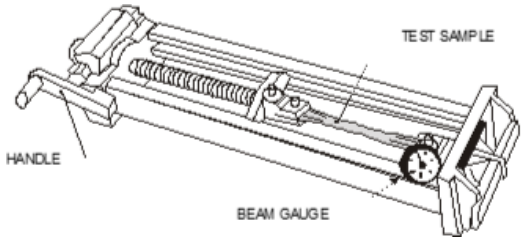
In order to discover the various properties of a material we must carry out material tests. There are many different types of tests available but the most common is the tensile test. As the name suggests the material is subjected to a tensile force or in other words, the material is stretched or pulled apart.

Results from tensile tests allow us to determine the following properties:

1. The elasticity of a material
2. The plasticity or ductility of the material
3. The ultimate tensile strength of the material.

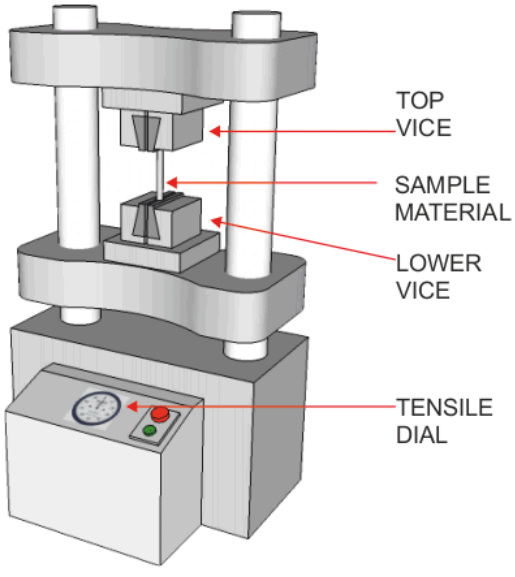
A tensometer or tensile testing machine is designed to apply a controlled tensile force to a sample of the material. A tensometer that might be found in schools is shown below.



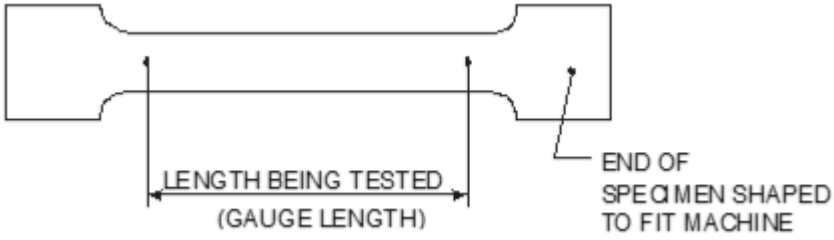


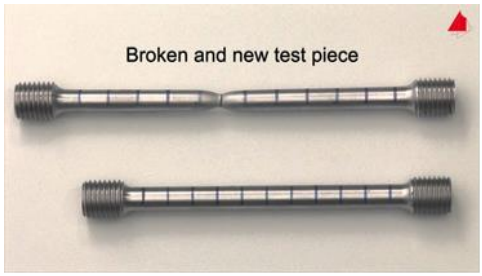
More sophisticated tensometers are available and are commonly used in industry. The main advantage of these machines is that they are able to plot a graph of how the material behaves during the test.

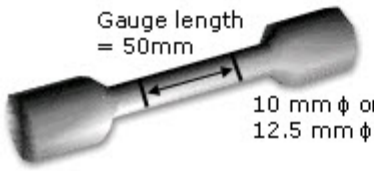


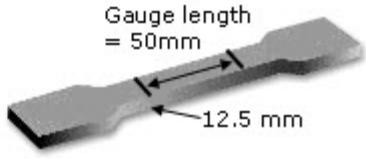


In order for tests to be carried out on a consistent basis, the shape of the specimen to be tested must conform to British Standards. The test sample is prepared to have a thin central section of uniform cross-section. A typical test specimen is shown below.









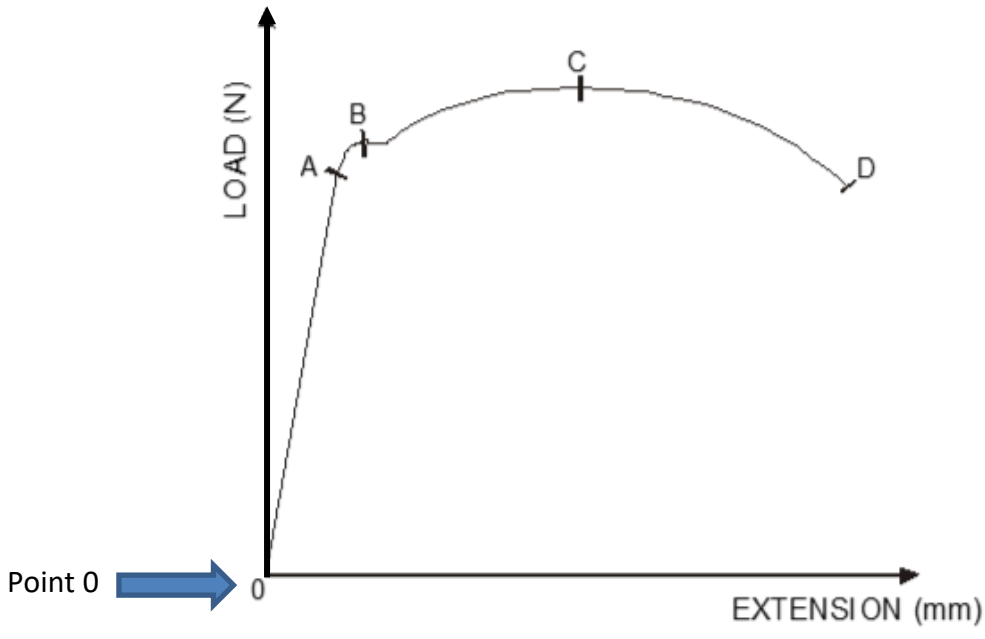
The principle of tensile testing is very simple. As the force is applied to the specimen, the material begins to stretch or extend. The tensometer applies the force at a constant rate and readings of force and extension are noted until the specimen finally breaks. These readings can be plotted on a graph to show the overall performance of the material.

The results of a typical tensile test for a sample of mild steel

Load V's Extension graph

The shape of the graph is very important to engineers and is used to predict how the material will behave or react under different loading conditions and different forces. The axes have been labelled as Load (force N) V's Extension (mm). Study the graph below as there are 5 important area that must be identified.

Yield point Elastic range Ultimate load Breaking point Plastic range



Tensile test on a specimen of mild steel

Between points **0** and '**A**' the material behaves elastically and this part of the graph is known as the **elastic region**. This means that the material stretches under the load but returns to its original length when the load is removed.

In fact, the Load (N) and extension (mm) produced are proportional and this part of the graph will be a straight line. This relationship is known as Hooke's Law and is very important to structural engineers.

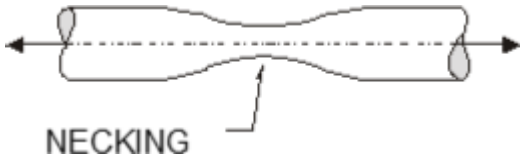
Point '**A**' is called the **Limit of Elasticity**. Any loading beyond this point results in plastic deformation of the sample.

Point '**B**' is called the **yield point**, and a permanent change in length results even when the load is removed. Loading beyond this point will result in a rapidly increasing extension.

The area between points '**B**' and '**D**' is known as the plastic range. This is so called because the material behaves in a plastic or ductile manner.

Point '**C**' the maximum or **ultimate tensile force**. This is the point where the maximum load or force the material can withstand is reached.

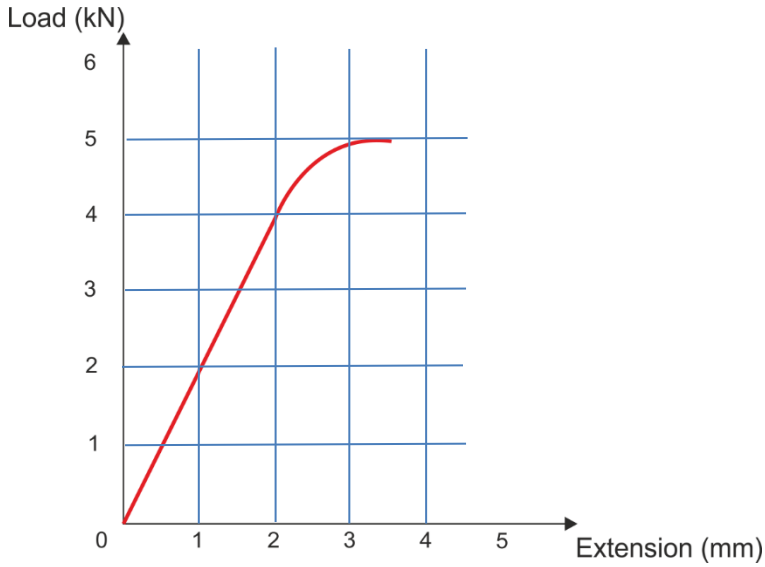
Between '**C**' and '**D**', the amount of force the material can now withstand deteriorates rapidly, and the cross-sectional area of the sample reduces or 'necks' (becomes thinner) and therefore reduces the strength of the sample.



The sample eventually breaks or fractures at point '**D**'. The shape of a typical fractured specimen is shown below.



A graph of a tensile test will allow us to read values of load and the amount the test piece has stretched in response. We can use these values to help us calculate the energy stored in the material at different points in the elastic and plastic ranges of the material.



Looking at the graph above, we can see that when a 4 kN load is applied the test piece stretches 2 mm. Using the equation for strain energy we can calculate how much is stored at that point.

$$E_s = ½ \times \text{Load} \times \text{extension}$$
$$= ½ \times 4000 \times 0.002$$
$$= \underline{\underline{4\ J}}$$

Assignments: Tensile Testing

1. You intend to carry out a tensile test on 4 pieces of materials: a basic aluminium alloy, Aluminium HE30, a mild steel alloy and PVC. It is to establish some of its physical properties and characteristics. You will be provided with a tensile tester equipment 'ES6', user guide and material sample worksheets.

a) Explain, briefly, how you would carry out such a test and what equipment would be required to do this.

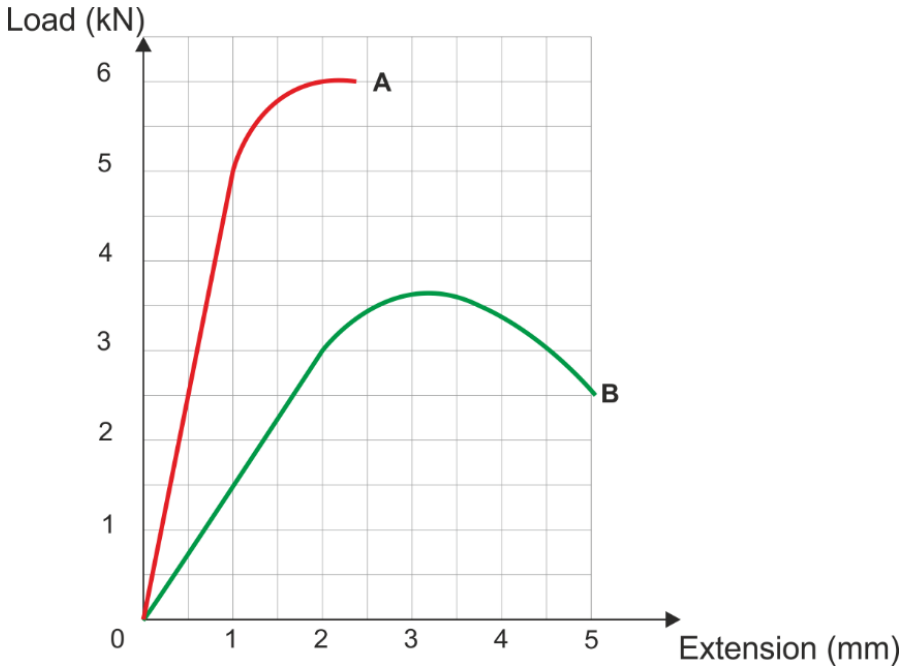
b) Sketch a typical specimen test piece that would be used in such a test. Indicate clearly two important physical readings that would have to be taken at the beginning of the test.

c) State one property that may be established from the results of the test.

2. On the graphs in the worksheets provided, plot a typical load v extension graph for all material specimens that have undergone a tensile test to destruction. On each graph, indicate the following points:

Yield point Elastic region Plastic region Breaking point Maximum load

3. The diagram below shows the results from tensile tests on two identically shaped test specimens made from two separate materials.



a) State five properties that can be compared between the two.
b) Describe the properties of each material in relation to the other.
c) Calculate the energy stored in each material at its yield point.
d) Explain the difference between plastic deformation and elastic deformation with respect to engineering materials.

4. Explain the difference between plastic deformation and elastic deformation with respect to engineering materials.

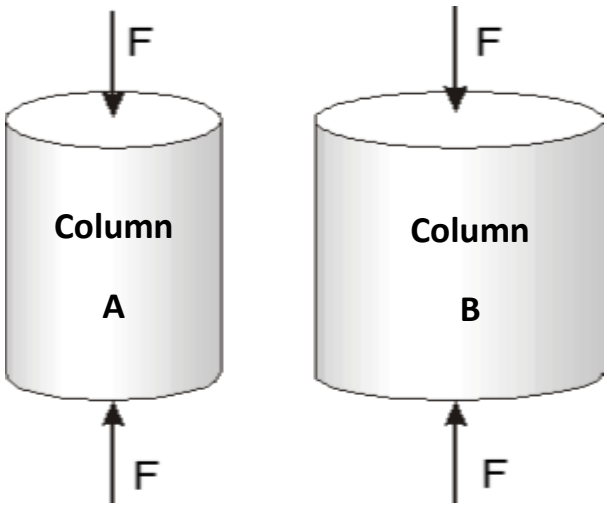
5. Give an example of two pieces of modern engineering design: one designed to display elastic deformation; the other to display plastic deformation.

Stress on members

When mechanical or structural engineers are working on a project, they need to select the right materials (wood, steel, iron, plastic etc). Two things they need to know are whether the materials they choose will be strong enough and whether they will stretch or compress too much when forces are applied.

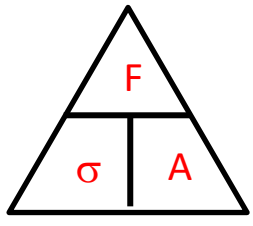
When a force is applied to a part of a structure, it will experience **STRESS**. If the level of force which is applied to the member is too high then the stress which is created may cause the part to bend, buckle or worst case, fail and break. Far more useful to an engineer than a load extension graph is a stress strain graph. This in many ways resembles a load extension graph but the data in this form can be interpreted more easily in design situations. First let us examine what is meant by stress and strain.

The stress can vary by the force put on it, but also by the size of the structure. For example; if the same load is applied to column A and column B which are of the same length, then Column A will suffer **more** from Stress and will be more likely to fail. The reason why is because Column A has a far smaller cross-sectional area.



Where Force is measured in Newtons (N) and the Area is the cross-sectional area measured in mm^2. Stress therefore is measured in Nmm^{-2} (the unit measurement can also be given as N/mm^2). This means the amount of force acting on a structure is Newtons '**per**' square mm). It is denoted by the Greek letter Sigma (σ), it is calculated using the formula:

$$\text{Stress} = \frac{\text{Force}}{\text{Area}} \qquad \sigma = \frac{F}{A}$$



Teacher worked examples: Copy the questions and answer if required in your jotter.

1) A square bar of 20mm x 20mm cross section is subjected to a tensile load of 500N. Calculate the stress in the bar.

$$Stress = \frac{Force}{Area}$$

$$\sigma = \frac{F}{A}$$

$$\sigma = \frac{500}{400}$$

$$\sigma = 1.25 N/mm^2$$

Stress in the bar = 1.25 N/mm^2

2) A wire 4mm in diameter is subjected to a force of 300N. Find the stress of the wire.

Stress = Force ÷ Area **(Area= ∏ r^2 where the radius is 2mm)**

= 300 ÷ 3.14 x 2^2

= 300 ÷ 12.56 **= 23.9 N/mm^2**

3) The stress in a steel wire supporting a load of 8kN should not exceed 200N/mm^2. Calculate the minimum diameter of wire required to support the load.

Stress = Force ÷ Area

Area = Force ÷ Stress

Area = 8000 ÷ 200

Area = 40mm^2

Area = ∏ r^2 or $\frac{\pi d^2}{4}$

r^2= Area/∏ **Therefore** $d = \sqrt{\frac{4xA}{\pi}}$

= 40 ÷ 3.14

r^2= 12.74

r = √12.74

r = 3.57

d = 7.14mm

Minimum diameter of wire required to support load = 7.14 mm

4) A column of section 0.25m^2 is required to act as a roof support. The maximum allowable working stress in the column is 50 MNm^{-2}. Calculate the maximum compressive load acting in the column.

$$Stress = \frac{Force}{Area}$$

$$Force = Stress \times Area$$

$$Force = 50 \times 0.25 \times 10^6$$

$$Force = 12.5MN$$

5) The figure shows an oil platform used in the North Sea. The platform is designed so that each of the three legs always carries an equal share of the **120 MN** load. The legs are **15m** in diameter and are made of steel reinforced concrete. Calculate the stress that is applied on **each leg.**



Stress = Force ÷ Area (∏ r^2)
Stress = 40MN ÷ 3.14 x 7.5^2
Stress = 40MN ÷ 3.14 x 56.25
Stress = 40MN ÷ 176.6
Stress = 0.227 MNm^{-2} or 226 kNm^{-2}

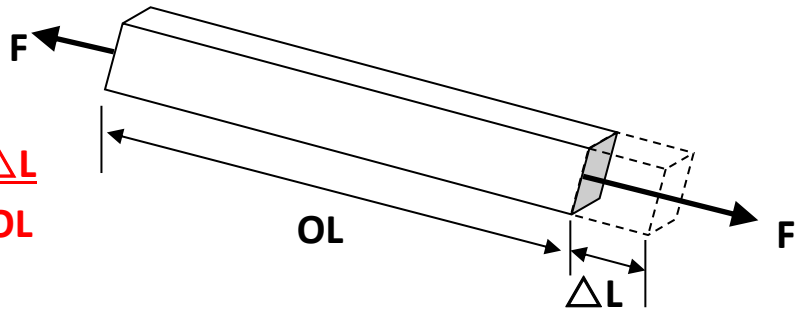
Strain on members

The result of applying a load or force to a structural member is a change in length. Every material changes shape to some extent when a force is applied to it. This is sometimes difficult to see in materials such as concrete and we need special equipment to detect these changes.

If a *compressive load* is applied to a structural member, then the length will reduce. If a *tensile load* is applied, then the length will increase. The result of applying a load to a structural member in **either** direction is called **STRAIN**. △L means the '*change*' in length when compared against the original OL.

Strain is calculated using the formula:

Strain = Change in Length $\varepsilon = \frac{\triangle L}{OL}$
Original Length



Where length in both cases is measured in the same units (m or mm). As the units cancel each other out, **strain is dimensionless**. This means that there are **no units of strain**. Put simply, strain is a ration that describes the proportional change in length in the structural member when a direct load is applied. Strain is denoted by the Greek letter *Epsilon* (**ε**).

Worked examples: Strain

A steel wire of length 5 m is used to support a tensile load. When the load is applied, the wire is found to have stretched by 2.5 mm. Calculate the strain for the wire.

$$\varepsilon = \frac{\Delta L}{L}$$

$$\varepsilon = \frac{2.5}{5000}$$

$$\varepsilon = 0.0005$$

Strain in the wire = 0.0005

Worked example: Strain

2. The strain in a concrete column must not exceed 5 x 10-4. If the column is 3 m high, find the maximum reduction in length produced when the column is loaded.

$$\varepsilon = \frac{\Delta L}{L}$$

$$\Delta L = \varepsilon \times L$$

$$\Delta L = \left(5 \times 10^{-4}\right) \times 3000$$

$$\Delta L = 1.5mm$$

Reduction in length of column = 1.5 mm

6.

Answer the following strain questions in your jotter.

a) A steel wire 4m long is used to support a load. When the load is applied the wire stretches by 2.5mm. Calculate the strain of the wire.

b) A wire 10m long stretches 5mm when a force is applied at one end. Calculate the strain.

c) The allowable strain on a bar is 0.0075 and its length is 2.5m. Calculate the change in length.

d) During testing, a steel rod stretches 0.6mm. If the resulting strain was 0.00012, what is the original length of the rod?

7.

a) A bar of steel 500 mm long has a cross-sectional area of 250 mm^2 and is subjected to a force of 50 KN. Determine the stress in the bar.

b) A wire 4 mm in diameter is subjected to a force of 300 N. Calculate the stress in the wire.

c) What diameter of round steel bar is required to carry a tensile force of 10 KN if the stress is not to exceed 14.14 N/mm2.

d) A wire 10 m long stretches 5mm when a force is applied at one end. Calculate the strain produced.

e) A tow bar, 1.5 m long, is compressed by 4.5mm during braking. Calculate the strain.

f) The allowable strain on a bar is 0.0075 and its length is 2.5 m. Calculate the change in length.

g) During testing, a shaft was compressed by 0.06 mm. If the resulting strain was 0.00012, what was the original length of the shaft?

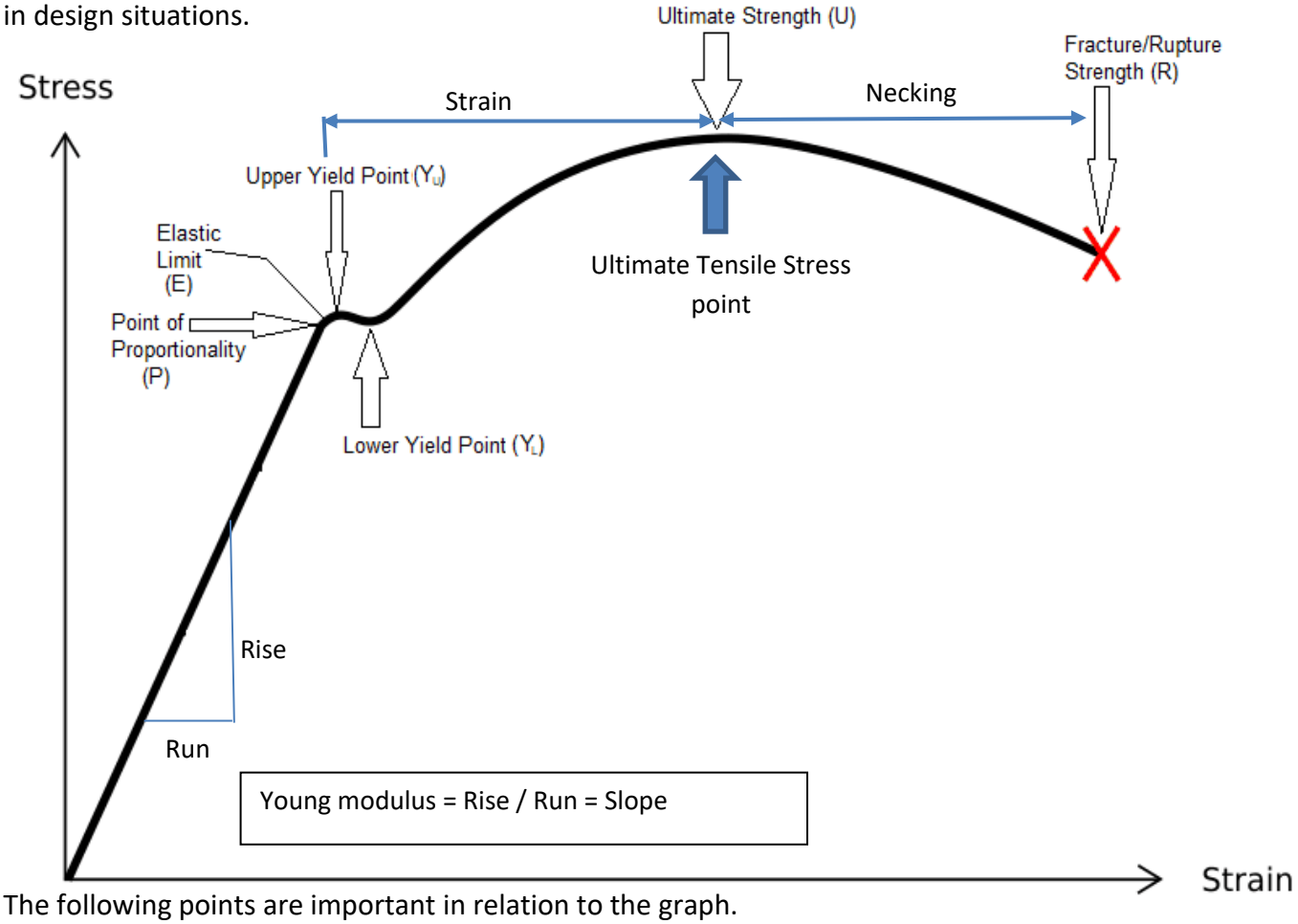
h) A piece of wire 6m long and diameter of 0.75 mm stretched 24 mm under a force of 120 N. Calculate stress and strain.

i) A mild steel tie-bar, of circular cross-section, lengthens 1.5 mm under a steady pull of 75 kN. The original dimensions of the bar were 5 m long and 30 mm in diameter. Calculate the intensity of tensile stress in the bar and determine the strain.

j) A mass of 2500 kg is hung at the end of a vertical bar, the cross-section of which is 75 mm x 50 mm. A change in length in the bar is detected and found to be 2.5 mm. If the original length of the bar is 0.5 m, calculate the stress and strain in the bar.

Stress V's Strain graph

When the stress and strain of a material are discovered, a stress v Strain graph can be created to display the findings. Be careful of the axes labels as the graph looks very similar to the load v extension graph but it does provide material data which makes it easier for engineers to interpret the information more easily in design situations.



The following points are important in relation to the graph.

1. Yield Stress

The yield stress is the maximum stress that can be applied to a structural member without causing a permanent change in length. The loading on any structural member should never produce a stress that is greater than the yield stress. That is, the material should remain elastic under loading.

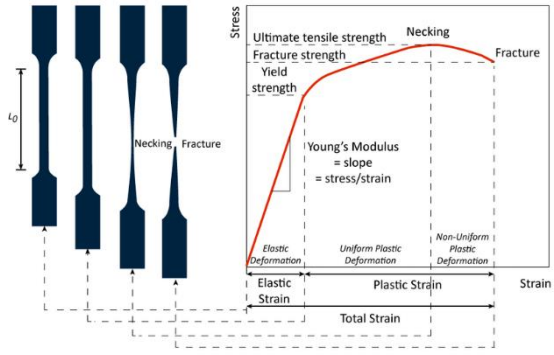
2. Yield Strain

The yield strain is the maximum percentage plastic extension produced in a material before it fails under loading. A ductile material such as copper needs to be formed and shaped into items such as pipes. For this to happen, it needs to be stretched to a point where it will no longer go back to its original shape, or deteriorate due to being stretched too much. For this to be effective, the material requires a high value of yield strain.

3. Ultimate Tensile Stress

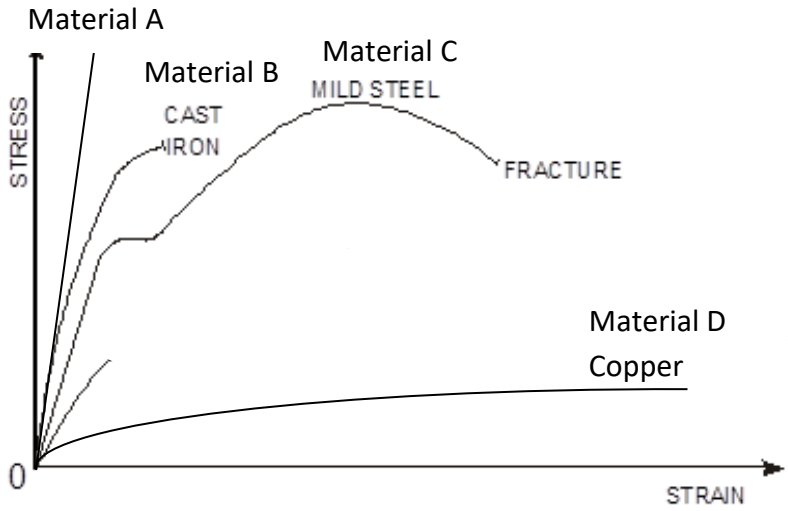
The Ultimate Tensile Stress (UTS) of a material is the maximum stress the material can withstand before it starts to fail. If a member in a structure is loaded beyond the UTS, the cross-section will reduce and the member will quickly fail.

Your teacher will provide you with a stress V Strain graph as shown which illustrates the change in samples during testing to be stuck in your jotter.



Using data from stress strain graphs

As we have already learned, vital information can be obtained from tensile tests when the data is plotted in the form of a stress strain graph. The graph opposite represents the relationship between stress and strain for common materials.



Types of materials

The graph above illustrates that material A has very little strain for the high amount of stress it can withstand. This means that it is very strong, but it is also quite brittle. This means its fracture will be a sudden action with little or no plastic deformation. (Glass or cast iron).

Material B is a strong material that is not very ductile. It has a plastic region, but this is small. This means it will stretch, but not in a huge amount before it breaks suddenly.

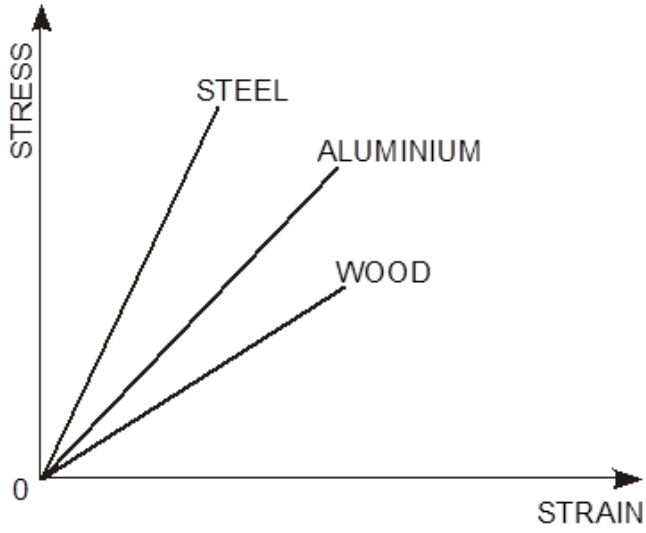
Material C is a ductile material as its strain is relative to the amount of stress it is put under. The material has a large elastic range and the plastic range is clearly visible in comparison to the other samples. The ultimate load is also noticeable and this is where it will start to deform and deteriorate before it hits the breaking point.

Material D has a very small elastic region but a large plastic region. It cannot take a huge amount of stress, but has a large strain. This means it will deform when a small load is placed on it, and it must have a high plasticity.

Youngs modulus

When a material is constantly loaded past its elastic limit, its performance becomes unpredictable. This could be disastrous, even fatal, if we consider the scale and type of structures we use every day. For this reason, structural engineers must ensure that projected stresses in structural members are held within the materials elastic limit.

When we test a range of common material we find that they all behave in an elastic manner up to a certain level of loading, even very brittle materials.



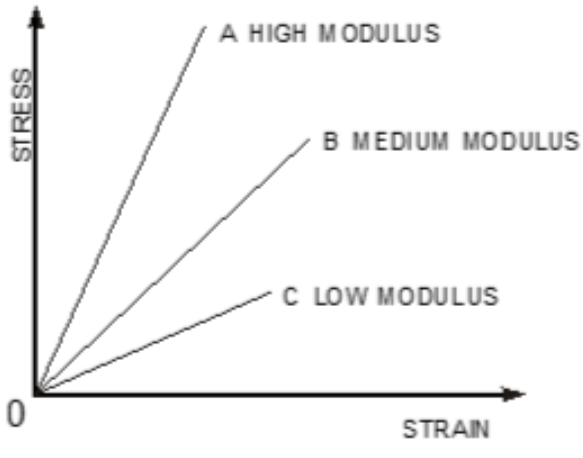
We also find that within the elastic limit, the graphs are a straight line therefore conforming to Hooke's Law. This means that stress is proportional to strain. We use the principle of Hooke's Law to find a value called young's Modulus. Young's Modulus is sometimes called the Modulus of elasticity and is calculated using the formula:

$$E = \frac{Stress}{Strain}$$

- measured in kN/mm^2

$$E = \frac{\sigma}{\varepsilon}$$

For any material, which obeys Hooke's Law, the slope of the straight line within the elastic limit can be used to determine young's Modulus. Although any value of stress and strain can be taken from within this region, it is customary for values to be taken from the graph at 50% of yield stress.



Modulus of elasticity determines the stiffness of a material. The higher the modulus, the greater the stiffness. Stiffness is a measure of a materials resistance to buckling under compressive loading. If a structural member starts to buckle it will bend and eventually collapse.

Worked example: Young's modulus

An aluminium tie rod is 1.5 m long and has a square cross-section of 20 mm x 20 mm. A tensile load of 5.6 kN is applied and produces a change in length of the rod of 0.3 mm. Calculate young's Modulus for the rod.

$$Young'sModulus = \frac{Stress}{Strain}$$

a) Calculate the stress in the rod.

$$\sigma = \frac{F}{A}$$

$$\sigma = \frac{5600}{40 \times 40}$$

$$\sigma = 14N/mm^2$$

b) Calculate the strain in the rod.

$$\varepsilon = \frac{\Delta L}{L}$$

$$\varepsilon = \frac{0.3}{1500}$$

$$\varepsilon = 0.2 \times 10^{-3}$$

c) Calculate Young's Modulus

$$E = \frac{\sigma}{\varepsilon}$$

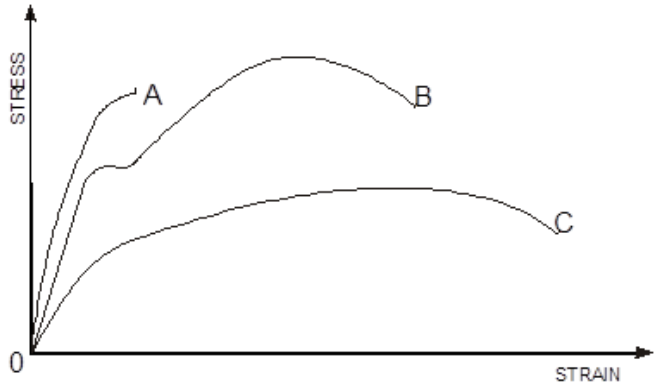
$$E = \frac{14}{0.2 \times 10^{-3}}$$

$$E = 70kN/mm^2$$

Young's modulus = 70kN/mm^2

Assignments: Young's Modulus

8. Image below shows a stress strain graph for three metals A, B and C.



a) Which metal is the most ductile?

b) Which metal can withstand the greatest load?

c) Which metal is the most elastic?

d) Which metal has the lowest modulus of elasticity?

e) Which metal is the stiffest?

f) Which metal is the most brittle?

9. A column 6 m high is compressed 1.5 mm when a load is applied. The column has a cross-sectional area of 7500 mm^2. Calculate the magnitude of the load if young's Modulus for the material has a value of 200 kN/mm^2.

10. A force of 1.5 kN is applied to the free end of a vertical wire 6 m long. The extension produced as a result was 12 mm. If young's Modulus for the material is 196 kN/mm^2, what is the diameter of the wire?

11. A test on a duralumin test piece 150 mm^2 in cross-sectional area and 50 mm long showed that it followed Hooke's Law up to a load of 38 kN with an extension of 0.16 mm for that load.

Calculate the stress at this load and the value for Young's Modulus for duralumin. Use your results to find the load that can be carried by a duralumin tension member 425 mm^2 in cross-sectional area and 325 mm long if the maximum permissible extension is 7.5 mm.

12. A bar of steel 500 mm long has a cross-sectional area of 500 mm^2 and is subjected to a direct pull of 20 kN. Determine the stress in the bar and the extension produced if Young's Modulus is 200 kN/mm^2.

13. A round brass rod in tension carries a load of 40 kN. Determine its diameter if the stress is not to exceed 40 N/mm^2. What will be the extension of the rod if the length is 2 m, and the modulus of elasticity for brass is 100 kN/mm^2?

14. A metal bar of length 250 mm and cross-sectional area 150 mm^2 is gradually loaded in tension up to the elastic limit. From the load-extension graph it is found that a load of 25 kN produces an extension of 0.2 mm. Calculate Young's Modulus of the material.

15. If a specification for steel in a structure limits the working stress for the material to 150 N/mm^2. Calculate a suitable diameter of a tie-rod if it requires to withstand a pull of 170 kN. Calculate the extension in the rod if the original length is 3 m and young's Modulus is 200 kN/mm^2.

16. A polypropylene rope is used to tie a small boat to a pier. The boat exerts a force of 750 N on the rope, which has free length of 6 m. If the rope should not stretch more than 30 mm, calculate the required diameter of rope.

Factor of safety

Most structures are extremely safe and well designed but due to unforeseen circumstances some structures fail or collapse. A structural engineer can never be absolutely certain that he/she has accounted for every possible type of load that will affect the structure. When a structure has failed, an investigation normally takes place to discover the reason for failure. The most common reasons include:

Overloading

This is when the load on the structure exceeds the value that was used during the design process. This type of failure may be due to the structure being used inappropriately, e.g. a man riding a child's bike, or because the circumstances have changed since the original design. This may be the case in bridges that were designed many years ago, where original calculations accounted mainly for cars passing over the bridge. Nowadays, we are only too aware of the increasing traffic on the road especially heavy goods vehicles. This leads to traffic jams where the traffic comes to a standstill. This may overload the structure beyond its design limit.

The most dangerous cause for a sudden change in loading on a structure is probably the weather. This is because of its unpredictable nature. No one can predict with any certainty what the weather will be like tomorrow, next week or next year. Freak weather conditions like hurricanes produce an additional force on a structure over and above what may have been calculated for a normal windy day.

Material/Joint failure

The material within the structure may fail if it is not of consistent quality or because it has deteriorated due to the working environment of the structure. We could never guarantee the performance of natural materials such as wood as they all contain natural defects such as knots, shakes etc. Some materials are susceptible to particular conditions, for example wood swells up as it absorbs moisture, mild steel rusts due to oxygen and water.

The joints used within the structure may fail because they are inappropriate and cannot support the load, or if they have been poorly made. This is particularly relevant with techniques such as welding. The welds on large structures are usually x-rayed in order to detect any defects.

Fatigue

It is difficult to predict exactly when a structure will fail. Repeated loading and unloading of a structure will wear down the material's resistance to breaking and eventually it will fail. This may even be the case if the load remains within the maximum used in the original design calculations. The principle of fatigue can be demonstrated by bending a paper clip backwards and forwards. The paper clip will not snap the first time, or probably the second. After that we are unsure just when the paper clip will fail. Each time we bend it we are not applying any greater a force but eventually the paper clip snaps.

Applying a factor of safety

Depending on the performance criteria which a structure must meet; a factor of safety will be applied to the design. Factors of safety vary from one structure to another, depending on the consequences of failure. The factor of safety applied to a nuclear power station is much higher than that for a conventional power station because the implications of structural failure are far more serious. The factors of safety applied to any design is decided through the experience and knowledge of the designer in charge, as well as close examination of the structure itself and the job it is expected to do. The higher the value for factor of safety, the more cautious the engineer is about the design.

The actual load which the structure or component is designed to carry is only one factor in a complicated process. The designer must remember certain other things such as that very high quality materials are expensive; very accurate dimensions are difficult to achieve during manufacture; quality of bolts, rivets or welds may vary; there may be very high stresses during the construction process; and so on. The following points affect the decision on the factor of safety:

1. The value of the maximum load and accuracy of calculations.
2. The type of load on the structure.
3. The reliability/quality of the material.
4. The effect of corrosion or wear on the dimensions of the structure.
5. Errors during manufacture or construction.
6. The consequences of failure.

To help with this process, the designer might ask the following questions:

- What is likely to be the cause of structural failure? Have I considered every possibility even the obvious causes such as the material, shape of the structure, joining techniques?
- What are the implications if the structure fails? What damage will be caused and what effect will this have on human life and existence?
- What are the operating conditions of the structure like? Is this a harsh working environment in terms of weather or chemicals that may corrode the materials within the structure?
- What external factors or conditions might affect the structure? What is the structure built upon? Is there likely to be any 'freak', one-off incidents such as a sudden impact?
- Are there likely to be changes to any of these conditions within the working life of the structure?

It is easy to think that we can improve the performance of a structure by increasing the size or thickness of members within the structure, even increasing the number of structural members. In doing so however, it is possible to make the structure weaker and more susceptible to failure. As you make these changes, the structure becomes heavier and loading increases under its own weight. You may be solving one problem but creating another.

Calculating factor of safety

We can apply a factor of safety to a structure in one of two ways. The first is in terms of the loading the structure can withstand and the second is in terms of the stress within the structure.

$$Factor of Safety = \frac{Ultimate Load}{Safe Working Load}$$

$$Factor of Safety = \frac{Ultimate Stress}{Safe Working Stress}$$

If a structure is designed to support a load of say 10 kN and a factor of safety of 2 is applied to the design, then in fact the structure should be able to support 20 kN. In industrial settings such as factories, lifting devices are marked to show how much the weight can be lifted safely. This is known as the safe working load (SWL).

The loading on any structural member should never produce a stress that is greater than the yield stress of the material that the member is made from. In fact, the working stress to which a structural member will be subjected is generally well below the material's yield stress, therefore operating well within the elastic region. This ensures that if any sudden unexpectedly high loading is applied, the stress in the structural element will not exceed the yield stress preventing permanent deformation in the element. A factor of safety (FOS) of 2 means that the design is stressed to half the value at the yield point.

Worked example: Factor of safety

The maximum ultimate tensile stress for aluminium is 300 N/mm^2. If the working stress on a component is 50 N/mm^2, calculate the factor of safety applied in the design of the component.

$$Factor of Safety = \frac{Ultimate Stress}{Safe Working Stress}$$

$$Factor of Safety = \frac{300}{50}$$

$$Factor of Safety = 6$$

Accessing tabulated data

When designing a structure, an engineer would not be expected to test every piece of material that is used. The most commonly used materials have been tested exhaustively and the test data is available in British Standards publications. The table below is a copy of information provided within the SQA data booklet for use in exams.

Material	Young's Modulus E $kNmm^{-2}$	Yield stress σ_y Nmm^{-2}	Ultimate tensile stress Nmm^{-2}	Ultimate compressive stress Nmm^{-2}
Mild steel	196	220	430	430
Stainless steels	190–200	286–500	760–1280	460–540
Low-alloy steels	200–207	500–1980	680–2400	680–2400
Cast iron	120	–	120–160	600–900
Aluminium alloy	70	250	300	300
Titanium alloy	110	950	1000	1000
Nickel alloys	130–234	200–1600	400–2000	400–2000
Concrete	–	–	–	60
Concrete (steel reinforced)	45–50	–	–	100
Concrete (post stressed)	–	–	–	100
Plastic, ABS polycarbonate	2·6	55	60	85
Plastic, polypropylene	0·9	19–36	33–36	70
Wood, parallel to grain	9–16	–	55–100	6–16
Wood, perpendicular to grain	0·6–1·0	–	–	2–6

An engineer would examine and use this information to help work out what material should be used for any given job. By examining the table, some materials perform quite differently in tension and compression, for example soft brass performs better in compression than tension. Other materials perform equally well under both types of loading, for example low-alloy steel and titanium alloys.

For each material listed in the table, data is given on:

Young's Modulus - the elasticity or stiffness of a material.

Yield Stress - the value of stress which, if exceeded, will result in the material changing length permanently.

Ultimate Tensile Stress - the maximum value of stress the material can withstand before failing due to tension.

Ultimate Compressive Stress - the maximum value of stress the material can withstand before failing due to compression.

Example:

A supporting member for an extruding shop sign is in tension. When it is still in its elastic range and under a load of 700N, it extends by 0.11mm. The support wire has a diameter of 10mm and a length of 2,500mm

a) Calculate Youngs Modulas for the support wire.
b) State the name of the material used in the support wire.

Answers:

a) $A = \pi r^2$ $3.14 \times 5^2 = 78.5mm^2$

$\sigma = F / A$ $700 / 78.5$ $= 8.9mm^{-2}$

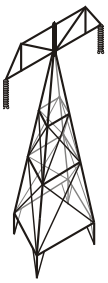
$\epsilon = \Delta L / L$ $0.11 / 2{,}500 = 0.000044$

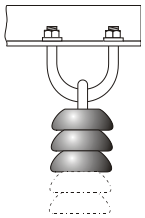
$E = \sigma / \epsilon$ $8.9 / 0.000044 = 202KNmm^{-2}$

b) Now that the Young's Modulas has been discovered, the material used can now be identified. By looking at the Youngs Modulas column on the SQA data table, the only material that fits into the range is a **low-alloy steel**, as its Young's Modulas is between 200 and $207KNmm^{-2}$.

Assignments: Tabulated data

17. Image shows a general view of an electricity pylon. The cross-arm supports conductor wires by means of an insulator suspended from its end.





The insulator above is fixed to the pylon by means of a low-alloy U-bolt shown below. Explain why a high factor of safety is required for this particular application. Suggest ways in which the structure might fail.

18. A lighting gantry in a theatre is suspended from the ceiling by wire ropes. The gantry hangs above the audience and is designed to hold 10 lights. A factor of safety of 10 is applied to the design. Explain the reasons why the engineer has used such a high factor.

19. In an Art Gallery, a sphere weighing 10 kN is to be suspended from the ceiling by a stainless steel bar 20 mm in diameter and 3 m long.

a) Select a value for Young's Modulus for Stainless Steel and hence calculate the extension of the bar when loaded.

b) Explain any assumptions made.

c) If a factor of safety of 5 was applied to the design, discuss the effect of this on the dimensions of the bar. (Use the Data Booklet.)

20. A steel overhead wire 25 mm in diameter has an ultimate tensile stress of 1250 N/mm^2 and a value for Young's Modulus of 207 kN/mm^2. If the factor of safety is 5, calculate the allowable pull on the wire and find the corresponding change in length on a 36 m span.

21. The maximum load in a tensile test on a mild steel specimen is 76 kN. If the test piece is 15 mm in diameter, calculate the ultimate tensile stress. What is the working stress and greatest allowable load on a bar 30 mm in diameter made from the same material? Use a factor of safety of 3.

22. A tensile test on a specimen of an unknown material provided the following data: Cross-sectional area = 120 mm^2, Gauge (original) length = 75 mm, Load (within elastic limit) = 30 kN, Extension of gauge length = 0.17 mm

a) Determine Young's Modulus of the material.
b) By referring to the Data Booklet, suggest which material was being tested.
c) Calculate the factor of safety for this material for the conditions described by the data given above.

23. A stainless steel tie-bar in a structure is to be designed to carry a load of 450 kN, with a factor of safety of 4.

a) Explain why this material is suitable for a use as a tie but not a strut.
b) Calculate the diameter of the bar.
c) Calculate the amount the tie-bar will stretch if its original length is 3.5 m.

24. A mild steel bar 25 mm in diameter and 0.5 m long is held in tension as it supports a sign above a shop.

a) Find the safe load that can be supported by the bar if a factor of safety of 8 is applied to the design.
b) Find the extension in the bar under this load.
c) Describe ways in which the bar might fail given these conditions.

25. A trapeze wire is made from a material for which the ultimate tensile stress is 1000 N/mm^2 and the yield stress is 950 N/mm^2. The wire has a working load of 3 kN.

a) Calculate the factor of safety to keep the stress just within the elastic limit for the material.
b) Calculate the diameter of the wire for this factor of safety.
c) Calculate the diameter of the wire for a factor of safety of 3.
d) Which factor of safety would you apply to this design? Explain your answer.

26. If the ultimate tensile stress of a steel wire is 750 N/mm^2, what load could safely be carried by the wire of diameter 2.5 mm? Use a factor of safety of 5.

27. A single mild steel cable supports a lift. The safe working load of the lift is 30 kN and a factor of safety of 12 is applied to the cabling. If the length of the cable is 100 m and the allowable extension is 50 mm,

a) calculate the required diameter of the cable.
b) Explain why such a high factor of safety has been applied in this design.
c) Suggest one way to improve the design of this system.

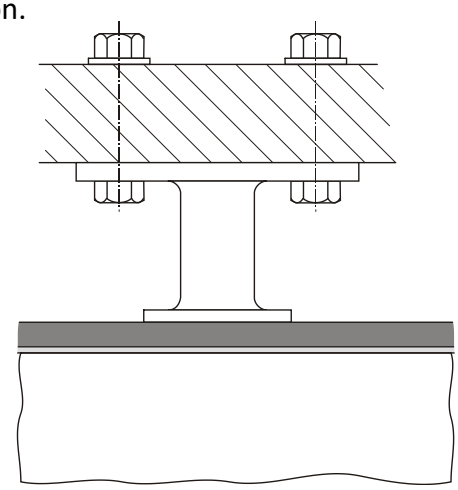
28. Winches are used on aircraft carriers to haul helicopters across the landing deck. The winch cables are made from mild steel and are fixed to the helicopter landing gear. The tension in the cables is 20 kN and a factor of safety of 8 is applied to the design. Calculate the diameter of the cable.

29. If the wire is 6 m long and the extension is 4.8 mm, what is the value of Young's Modulus?

30. A lifting jack has a maximum lift of 300 mm and the diameter of the ram is 75 mm. Find the maximum force the jack may exert if the ram has to have a factor of safety of 7 and the ultimate tensile stress of the metal is 483 N/mm^2. Find the change in length of the ram if the modulus of elasticity of the material is 201 kN/mm^2.

31. A tension member in a roof truss is subjected to a pull of 117 kN. The material has an ultimate tensile stress of 465 N/mm^2. When a factor of safety of 5 is used, find the diameter of the member. If the member is 3 m long and stretches 1.35 mm under loading, calculate the modulus of elasticity.

32. A display screen in a lecture theatre is held in place by a bracket fixed to the roof by two mild steel bolts as shown below. The bolts are initially tightened so that the tensile stress in them is 7 N/mm^2 before any load is applied. The weight of the screen is 3 kN and the bolts carry an equal share of this load. The bolts are designed with a factor of safety of 9. Determine the diameter of the bolts needed for this application.



33. Part of a pneumatic system is shown in figure 4. Compressed air is stored in the tank at a maximum pressure of 8 N/mm^2 and the 120 mm diameter outlet is sealed by means of a valve block which is held in place using 15 mm (M15) diameter bolts as shown. The normal stress set up in each bolt due to tightening is 6 N/mm^2 and the ultimate tensile stress for the bolts is 370 N/mm^2. The factor of safety for the system is 5.

a) Calculate the force on the value block using the formula, Force = Pressure x Area.
b) Calculate the safe working stress that can be carried by the bolts.
c) Calculate the stress in each bolt due to loading.
d) Calculate the load to be carried by each bolt.
e) Determine how many bolts are required to attach the valve block to the storage tank.

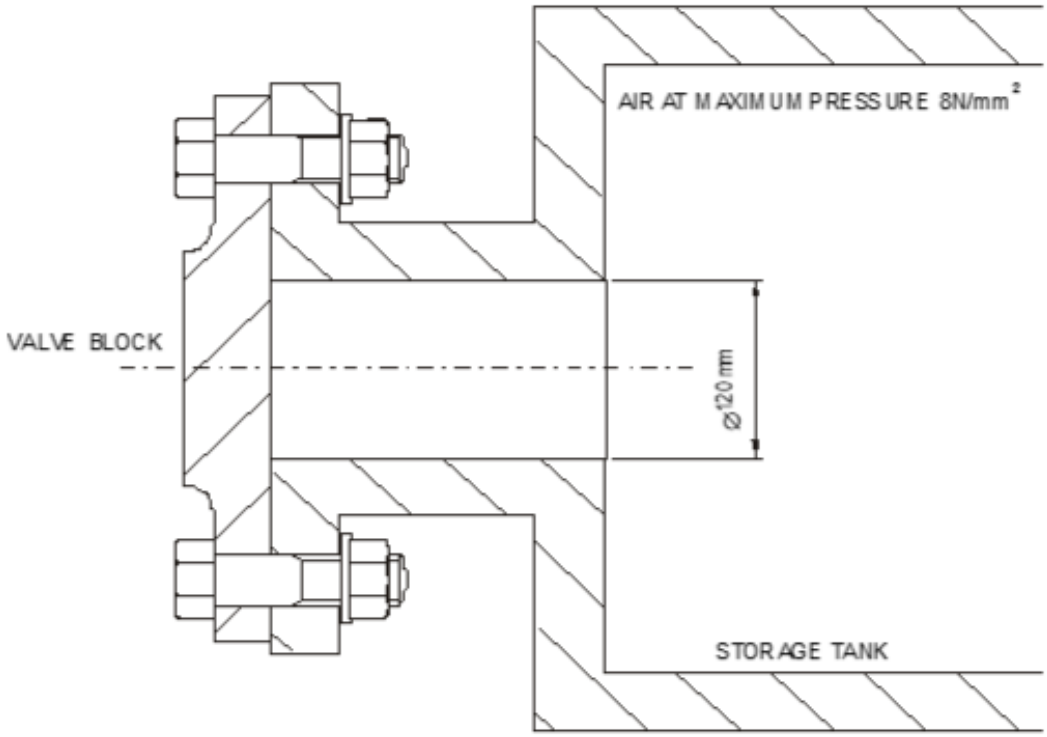


Figure 4

Exam questions

Marks

6. Figure Q6 shows the load-extension graph produced during a tensile test performed on an alloy-steel specimen.

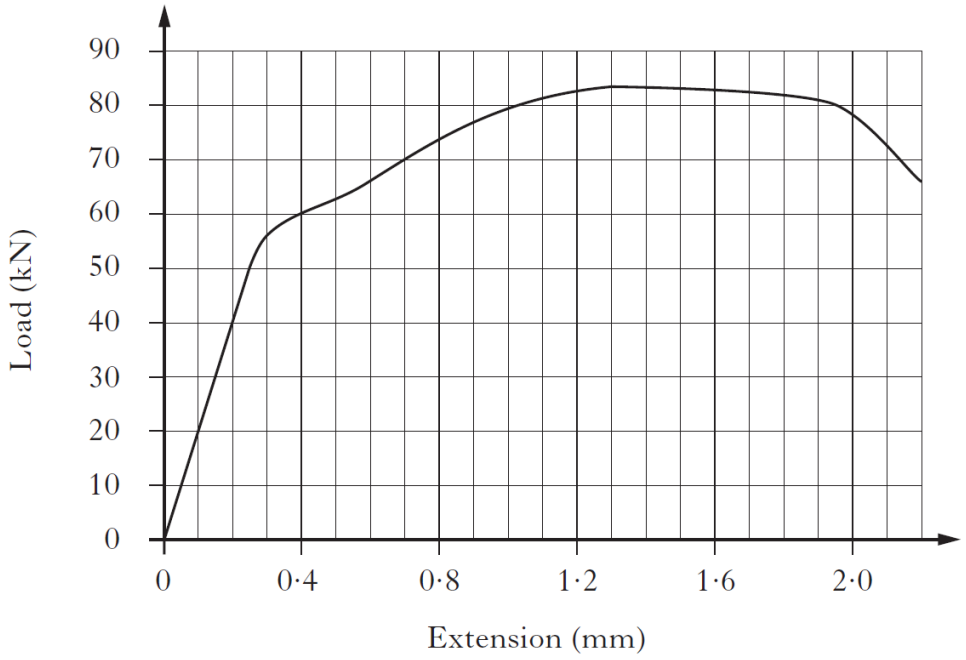


Figure Q6

The test specimen was 120 mm long with a rectangular cross-section of 26 mm × 6 mm.

(*a*) Calculate Young's Modulus for this material. **7**

(*b*) Describe the effect on the specimen of **applying** and then **removing** the following loads:

(i) 50 kN; **2**

(ii) 80 kN. **2**

(11)

Marks

2. A load-extension graph for a standard test specimen is shown in Figure Q2. The specimen is 200 mm long and 11·3 mm in diameter.



Figure Q2

(*a*) Calculate Young's Modulus for the test specimen. **5**

(*b*) State the name of the material from which the test specimen is made. **1**

(*c*) Sketch the graph shown in Figure Q2, and on it show the yield point, the ultimate load, the plastic range and the elastic range. **4**

Two further specimens were tensile tested. The dominant mechanical property of specimen A was brittleness, and that of B was ductility.

(*d*) Sketch, on the same axes, typical stress-strain graphs for specimen A and specimen B. Clearly label the axes and identify each graph. **4**

(14)

4. Figure Q4 shows a load-extension graph for a tensile test on a sample of a special type of steel.

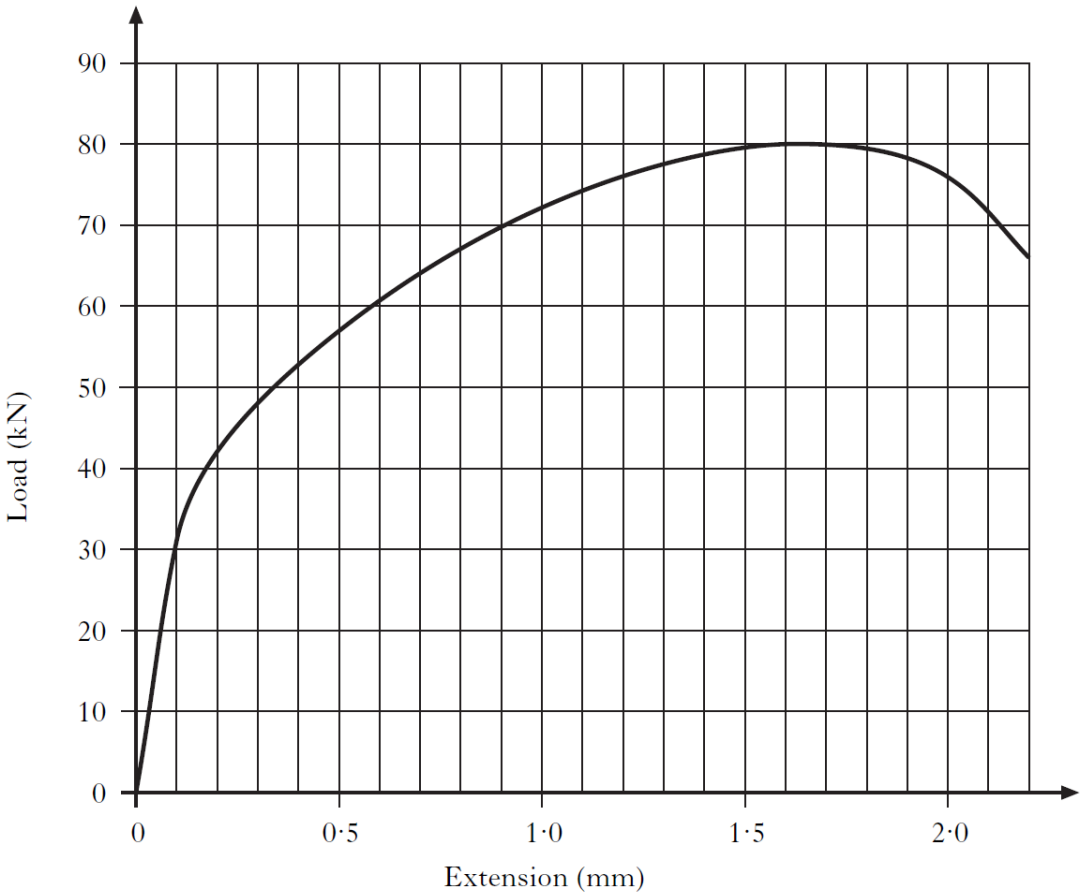


Figure Q4

(*a*) State whether this material is brittle or ductile.

The test specimen was 50 mm long with a cross-sectional area of 80 mm^2.

(*b*) Calculate Young's Modulus for the material.

(*c*) Calculate the **ultimate** tensile stress for the material.

A bar with a rectangular cross-section of 30 mm × 10 mm is manufactured from the material. The bar is to be loaded in tension with a Factor of Safety of 6.

(*d*) (i) Calculate the safe working stress for the bar.

(ii) Calculate the safe working load for the bar.

Spotlights are supported by a bracket at C as shown in Figure Q10(*b*). **Four** mild steel bolts support the bracket and are under tension.

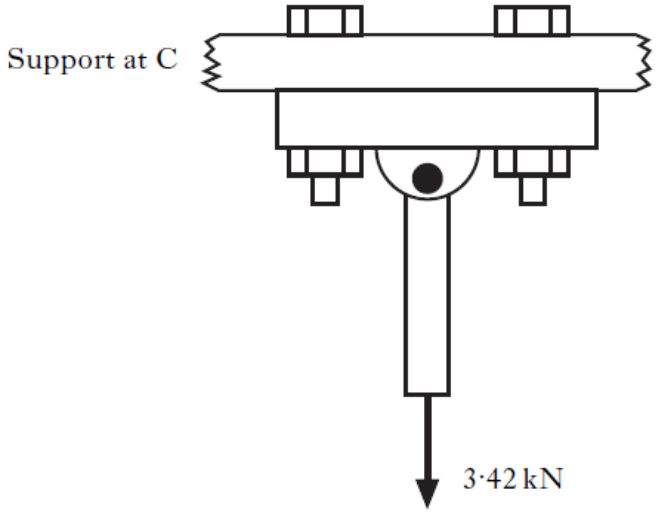


Figure Q10(*b*)

The tensile force in each of the four bolts due to tightening is 80 N, and the applied loading on the bracket is 3·42 kN. A Factor of Safety of 8 is applied.

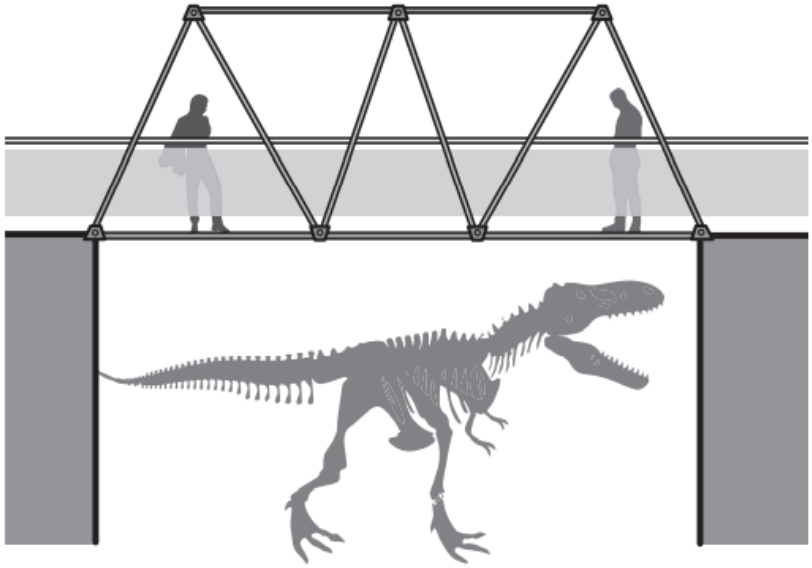
(*b*) Calculate the minimum diameter of each bolt. **5**

Q9 and Q10 2019 Higher

9. A team of engineers is asked to design a walkway over a dinosaur exhibit for a natural history museum.

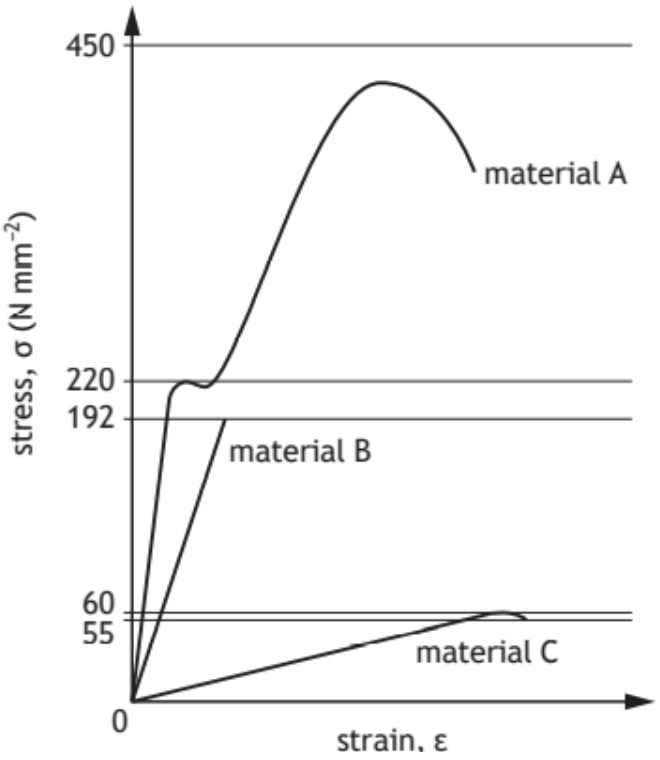


An initial design for a walkway over the top of the exhibit is shown below.



9. (continued)

Tensile test results for three materials considered for use in the walkway are shown in the stress-strain graph below.



(a) (i) Describe, with reference to the stress-strain graph, three **different** properties of material B in relation to material A or material C. **3**

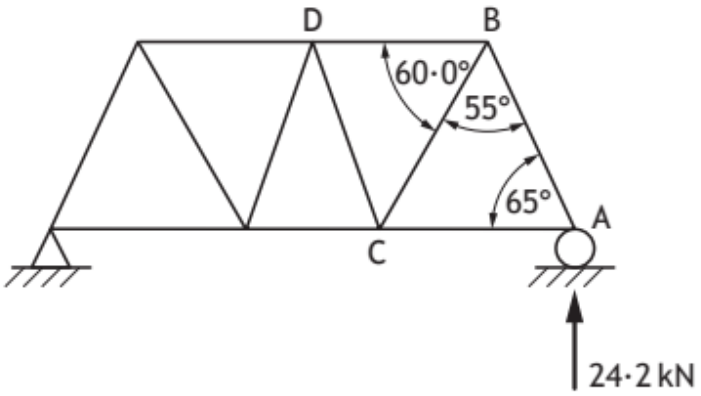
(ii) Identify, with reference to the Data Booklet, material A and material C. **2**

Material A _____

Material C _____

9. (continued)

A partially completed free body diagram of the design for the walkway is shown.



(b) Calculate, using nodal analysis, the magnitude and nature of the forces in members AB, AC, BC and BD.

Complete the table below. **7**

Show all working and final units on the page opposite.

Member	Magnitude	Nature
AB		STRUT
AC		TIE
BC		
BD		

MARKS

10. A mechanical engineer produced a preliminary design for a pneumatic system to open and close a dinosaur skeleton's jaw.



The pneumatic cylinder used to move the jaw has an 8·0 mm diameter **aluminium alloy** piston rod to support a load of 33·2 kN.

(a) (i) Calculate the factor of safety applied to the design of the piston rod. **4**

MARKS

10. (a) (continued)

(ii) Comment on the appropriateness of the factor of safety of the piston rod. 1

The mechanical engineer decides to use a different pneumatic cylinder to support the 33·2 kN load.

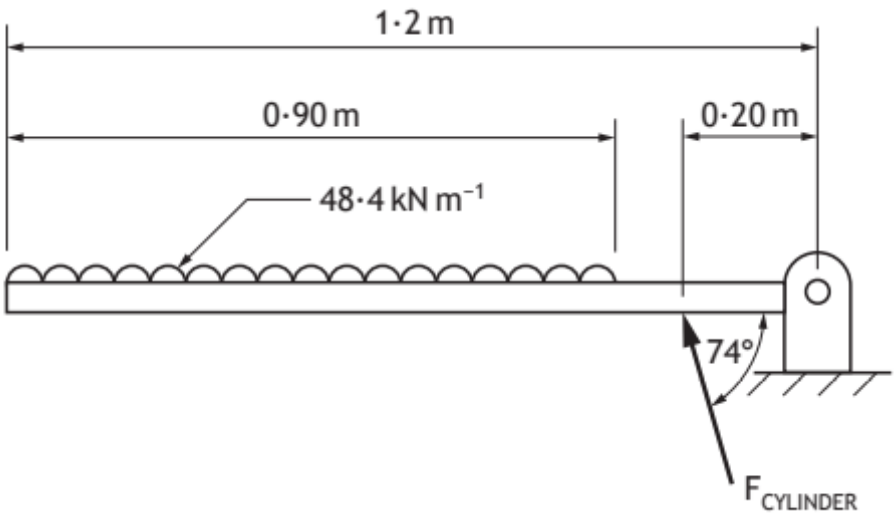
The piston rod area is 491 mm^2 and is made from **titanium alloy** with a length of 0·78 m.

(iii) Calculate the change of length in the piston rod under these conditions. 4

10. (continued)

After testing, the pneumatic cylinder is repositioned to move a hinged beam in the jaw of the dinosaur as shown below.

The uniformly distributed load of the jaw is 48·4 kN m^{-1}.



(b) (i) Calculate the magnitude of force $F_{CYLINDER}$. **3**

(ii) Calculate the magnitude and direction of the reaction at the hinge. **4**